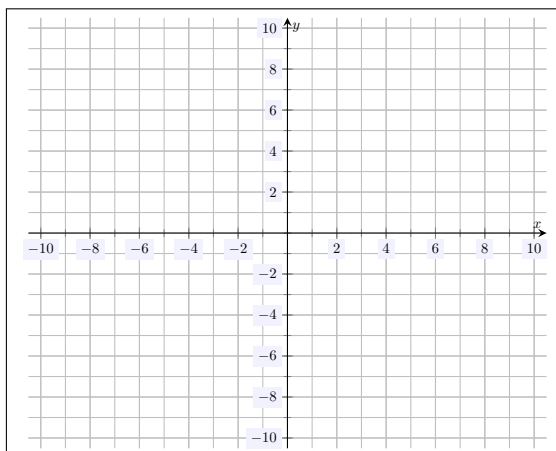
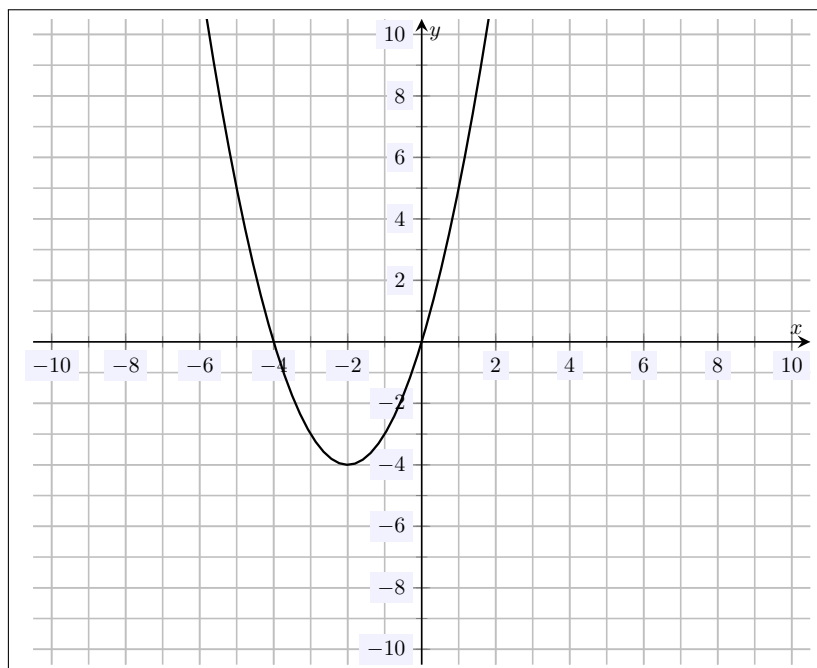


Recitation: Polynomials (Linear & Quadratic Functions)

Problem 1. Sketch the function $f(x) = 7 - (x - 5)^2$.



Problem 2. Find the equation of the quadratic function shown below. Be sure to fully justify why your answer is correct.



Problem 3. Solve the following equation:

$$5\sqrt{2}x + 8 = -3(1 - 2x)$$

Problem 4. Consider the quadratic function $f(x) = x^2 - 6x + 14$.

- (a) Find a, b, c for this quadratic function.
- (b) Does $f(x)$ open upwards or downwards? Explain.
- (c) Is this quadratic function convex or concave? Explain.
- (d) Find the minimum value of $f(x)$, if it exists. If it does not exist, explain why.
- (e) Find the maximum value of $f(x)$, if it exists. If it does not exist, explain why.